

Kasra Torshizi

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EDUCATION

University of Maryland, College Park

PhD in Computer Science

Jan 2025 - 2029

Advisor: Dr. Pratap Tokekar

University of Maryland, College Park

Master of Science in Computer Science

Aug 2023 – Dec 2024

3.89 GPA

University of Maryland, College Park

Bachelor of Science in Computer Science, Minor in Mathematics

Aug 2020 – May 2023

3.83 GPA

RESEARCH INTERESTS

Robotics, Reinforcement Learning, Imitation Learning, Optimal Control, POMDPs, Constrained Optimization, Diffusion Planners, Perception

PUBLICATIONS

[J.2] Ryan Lowe, [Kasra Torshizi](#), Lucas Wedland, Violet Mwaffo, Pratap Tokekar, Donald Costello. **Autonomous UAS Control for Unmanned Aerial Refueling Using DNN-Based Monocular Relative Position Estimation**. In submission to JIRS. *Equal Contribution

[J.1] [Kasra Torshizi](#)*, Chak Lam Shek*, Khuzema Habib, Guangyao Shi, Pratap Tokekar, Troi Williams. **Contextual Neural Moving Horizon Estimation for Robust Quadrotor Control in Varying Conditions**. In submission to IEEE RA-L. *Equal Contribution

[C.8] [Kasra Torshizi](#), Anukriti Singh, Sidharth Mathur, Khuzema Habib, Leo Du, Pratap Tokekar. **STeP: Signal Temporal Logic for Precise Specifications for Action Generation with Vision Language Models**. In submission to ICRA 2026.

[C.7] Khuzema Habib, Pranav Deshakulkarni Manjunath, [Kasra Torshizi](#), Troi Williams, Pratap Tokekar. **WESPR: Wind-adaptive Energy-Efficient Safe Perception and Planning for Robust Flight with Quadrotors**. IROS 2026.

[C.6] Anukriti Singh, [Kasra Torshizi](#), Khuzema Habib, Kelin Yu, Ruohan Gao, Pratap Tokekar. **AFFORD2ACT: Affordance-Guided Automatic Keypoint Selection for Generalizable and Lightweight Robotic Manipulation**. In submission to IEEE RA-L.

[C.5] Ryan Lowe, [Kasra Torshizi](#), Lucas Wendland, Violet Mwaffo, Donald Costello. **Autonomous UAS Control Leveraging DNN-Based Monocular Camera Relative Position Estimation for Unmanned Aerial Refueling Systems**. ICUAS 2026.

[C.4] [Kasra Torshizi](#), Donald H Costello, Pratap Tokekar. **Autonomous Drone Performance During Maritime Integration for Gust Response: Lab Experiment**. AIAA SciTech 2026.

[C.3] Chak Lam Shek*, [Kasra Torshizi](#)*, Troi Williams, Pratap Tokekar. **When to Localize?: A Risk-Constrained Reinforcement Learning Approach**. American Control Conference (ACC), 2025. *Equal Contribution

[C.2] Troi Williams, [Kasra Torshizi](#), Pratap Tokekar. **Where to Localize?: A POMDP Approach**. IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR), 2024.

[C.1] James Larisch, Waqar Aqeel, Michael Lum, Yaelle Goldschlag, [Kasra Torshizi](#), Leah Kannon, Yujie Wang, Taejoong Chung, Dave Levin, Bruce M Maggs, Alan Mislove, Bryan Parno, Christo Wilson. **Hammurabi: A Framework for Pluggable, Logic-Based x.509 Certificate Validation Policies**. ACM SIGSAC Conference on Computer and Communications Security, 2022.

EXPERIENCE

Reinforcement Learning Intern

Jun. 2026 – Oct. 2026

Systems and Technology Research

Woburn, MA

- Scaled deep reinforcement learning architectures by incorporating deeper networks, residual connections, and multi-skip connections to improve model capacity and optimization
- Evaluated weight initialization strategies to improve training stability and mitigate convergence to poor local minima
- Designed a supervised pretraining framework to initialize policies toward desired behaviors prior to reinforcement learning fine-tuning
- Developed a novel policy-regularization method to limit deviation from an imitation-learned policy while optimizing additional reinforcement learning reward objectives

Graduate Research Assistant

Nov. 2023 – Present

University of Maryland - The Robotics Algorithms and Autonomous Systems (RAAS) Lab

College Park, MD

- In-Progress Research: Developing a human-steerable MARL reward design pipeline that uses MPC guided demonstrations to infer temporal-logic constraints. These temporal-logic constraints are then constructed in reward machines for more aligned reward signals. Also investigating online policy adaptation with Residual MPC methods.

Graduate Research Assistant

Aug. 2025 – Present

The Maryland Autonomous Technologies Research Innovation and eXploration (MATRIX) Lab

California, MD

- Developing a control loop for autonomous UAV landings on Navy vessels, focusing on robustness in turbulent and low-visibility conditions
- Investigating decision-making methods for autonomous operation in partially observable environments, including visual degradation caused by water on the camera, reflections, and fog
- Work part of the MATRIX Lab Seed Grant
- Mentor: Dr. Donald Costello

Software Engineering Intern

May. 2023 – Aug. 2023

Echostar Corporation

Gaithersburg, MD

Undergraduate Research Assistant

May. 2021 – Nov. 2022

University of Maryland - Breakerspace Lab

College Park, MD

Software Engineering Intern

May. 2022 – Aug. 2022

Echostar Corporation

Gaithersburg, MD

GRANTS & FUNDING

- FY26 MATRIX Lab Seed Grant - \$75,000
- FY27 MATRIX Lab Grant - \$75,000

TEACHING (TA)

- CMSC351: Algorithms - Fall '22, Summer '24, Summer '25
- CMSC420: Advanced Data Structures - Spring '23
- CMSC421: Artificial Intelligence - Spring '25
- DATA602: Graduate Data Science - Fall '24

MENTORING

- Khuzema Habib - MS Robotics at UMD.
- Aarnav Kapoor - Undergrad CS at UMD.
- Leo Du - Undergrad CS at UMD.
- Suraj Modur - Undergrad CS at UMD. Now MS CS at Georgia Tech.

PRESENTATIONS

- 2026 AIAA SciTech
- 2025 Maryland Robotics Center Research Symposium (Lightning & Poster)
- 2024 Maryland Robotics Center Research Symposium (Poster)

CONFERENCE/JOURNAL REVIEWER

- CoRL ('26)
- ICRA ('26)
- IEEE RA-L ('26)
- IEEE SSRR ('24)
- IEEE Transactions on Machine Learning in Communications and Networking ('26)
- MRS ('26)

GRADUATE COURSEWORK

Computational Geometry, Decision-Making for Robotics, Computational Imaging, Numerical Optimization, Natural Language Processing, Human-Computer Interaction, Multi-Modal Foundation Models, Generative AI Agents

TECHNICAL FRAMEWORKS

Languages: Python, Java, C/C++, SQL, JavaScript, HTML/CSS, OCaml, Prolog, Bash

Libraries: Numpy, PyTorch, Matplotlib, SciPy, Pandas, OpenCV, Gym, Python Multiprocessing